Real-Time Video Generation With MiniMax H3

8× RTX PRO 5000 · vLLM-Omni

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Agenda

01 Toward Real-Time H3 Generation
02 vLLM-Omni and ComfyUI Applications

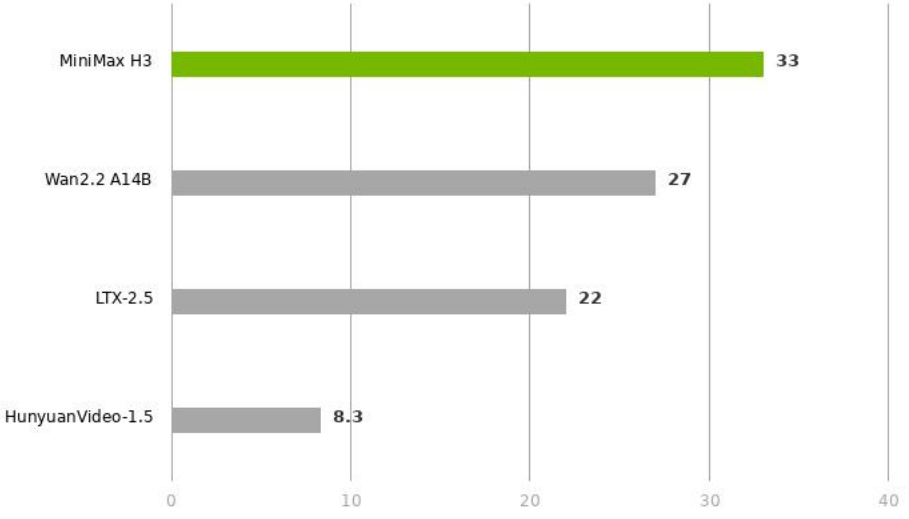


01 · Real-Time H3 Generation



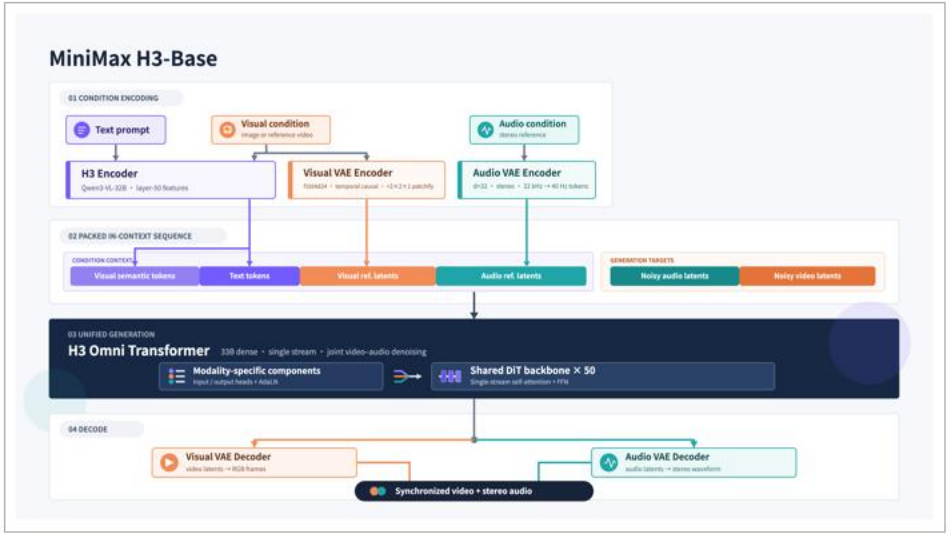
Real-Time Ref2VA Raises the Systems Bar

A 124-frame clip at 24 FPS is real time only when end-to-end latency stays below 5.167 s (RTF < 1)



Generative Backbone · Full Parameters (B)

RTF = latency ÷ media duration; the 124-frame deadline is 5.167 s. H3 has the largest dense backbone in this peer set; Wan2.2 has 27B total and 14B active parameters



Model Capability · Native 768P · 33B Dense

32B encoder → 33B dense Omni Transformer → visual/audio VAE. Fifty blocks × 50 steps keep compute and memory pressure high; about 13B AdaLN parameters can be cached

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 960×544 Ref2VA workload

SM120 Brings FP4 Compute; PCIe Limits Scale

All discrete SM120 cards pair fifth-generation Tensor Cores with GDDR7 and PCIe Gen5; none provides GPU-to-GPU NVLink

SKU	Compute	Memory	GPU I/O / Fabric
RTX PRO 5000 ×8 Blackwell	SM120 · 110 SMs 5th-gen Tensor · FP4	72 GB GDDR7 ECC 48 GB variant	PCIe Gen5 ×16 Planned: NIC → RDMA*
RTX PRO 6000 WS / Server	SM120 · 188 SMs 5th-gen Tensor · FP4	96 GB GDDR7 ECC	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	SM120 · 170 SMs 5th-gen Tensor · FP4	32 GB GDDR7	PCIe Gen5 ×16 No NVLink
DGX Spark · GB10	SM121 · 48 SMs 5th-gen Tensor · FP4	128 GB unified LPDDR5X	NVLink-C2C in SoC 200 Gb/s ConnectX-7

* Planned system path; GPUDirect RDMA requires a compatible NIC and PCIe topology · Sources: NVIDIA product specs, CUDA GPU table, GPUDirect RDMA guide

Latency Breakdown

NCCL occupies 49.2% of the mean GPU timeline

Stage	Mean Time per GPU*	Share
Mean GPU timeline*	250.0 s	100.0%
NCCL communication	123.0 s	49.2%
↳ DiT collectives	116.1 s	46.4%
↳ Encoder, VAE, and other collectives	7.0 s	2.8%
Tensor compute	84.5 s	33.8%
↳ Attention	45.3 s	18.1%
↳ GEMM / MLP	39.2 s	15.7%
AdaLN weight staging + device layout	38.5 s	15.4%
Encoder, VAE, and other GPU work	4.0 s	1.6%

- Of 38.5 s/GPU, 36.9 s is AdaLN projection weight staging from CPU; layout is 1.5 s
- This run uses the default runtime cache, not a complete precomputed schedule sidecar

Base Ref2VA · 960×544 · 124f @ 24 FPS · 50 steps · BF16 · 8× RTX PRO 5000 · default AdaLN offload · local NCCL / RDMA off · *mean across 8 GPU timelines

How Denoising Works

H3 follows a learned velocity field from noise toward clean video and audio latents

State and direction

- x(t) is the current latent; the DiT predicts a data-ward velocity

Solver update

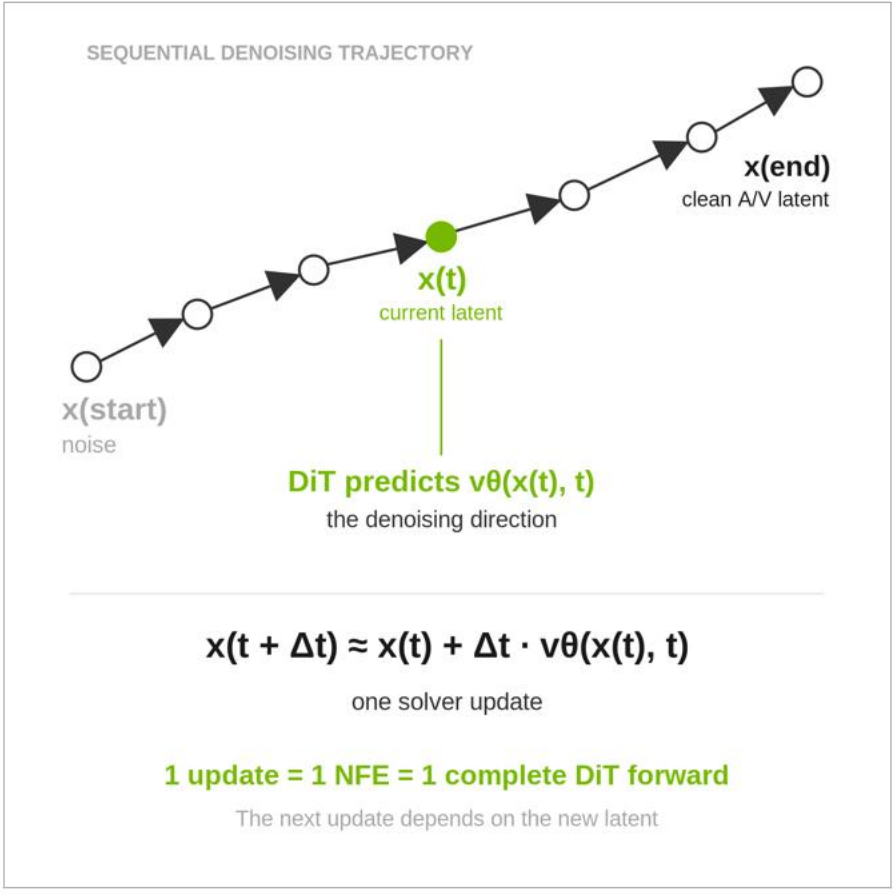
- Euler advances x(t) by a small interval along that velocity
- The next DiT call needs the new latent, so conventional steps run in sequence

Network function evaluation (NFE)

- One NFE is one complete DiT forward

Base H3

- 50 NFE × 50 main blocks = 2,500 main-block passes



Classical Flow-Matching Sampling

A chain of dependent solver updates turns noise into clean latents

Sources: MiniMax H3 Base configuration · NVIDIA FastGen-PDD sampling background (arXiv:2607.26004) · author abstraction

PDD Turns 50 DiT Passes Into 8

Predict four denoising directions in parallel; advance one interval block per pass

Teacher targets

- Base H3 supplies the mean denoising direction for each fine interval

One student pass

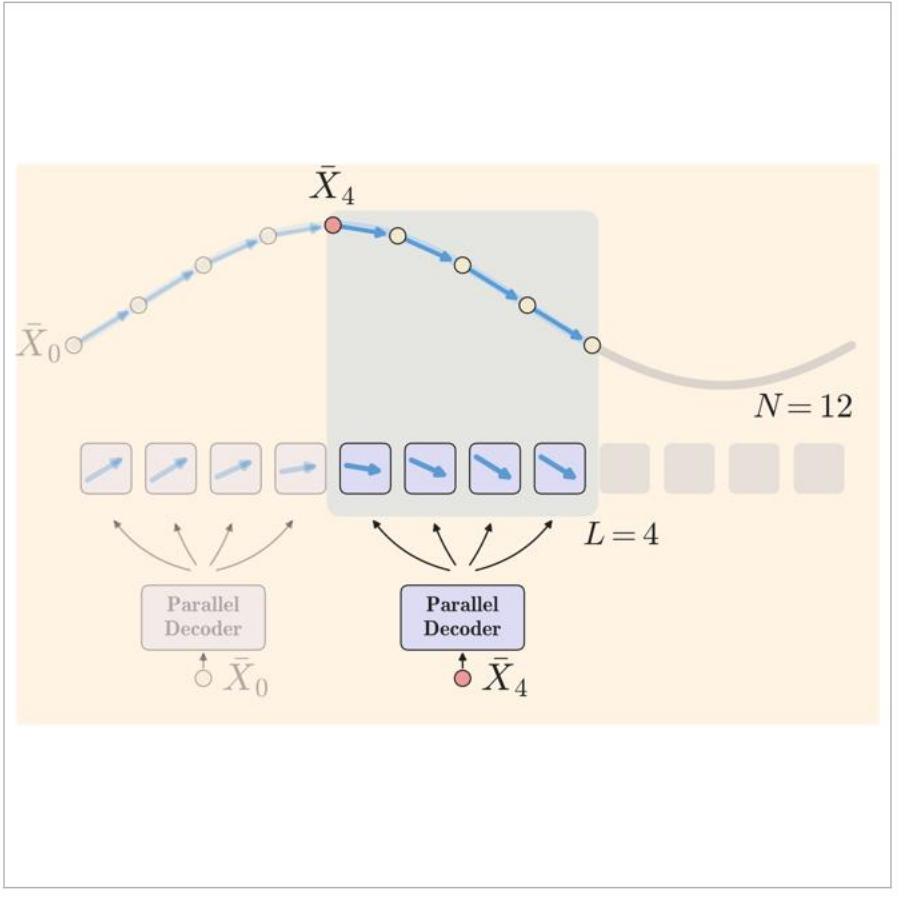
- From the same noisy latent, predict four consecutive directions in parallel

Advance and repeat

- Combine the four predictions to cross one block; repeat from the new latent

H3 PDD8

- Train a new 32-interval grid: 32 ÷ 4 = 8 passes
- Repeated block calls: 2,500 → 400 (−84%)



N = fine intervals · L = intervals predicted in parallel

Paper example: N = 12, L = 4 · H3 PDD8: N = 32, L = 4

Sources: NVIDIA FastGen-PDD, Fig. 2 (arXiv:2607.26004) · Alibaba-PAI MiniMax-H3 Ref2VA Acc-8Step @ 335001f

VSA Makes Attention Sparse

Video Sparse Attention learns which 3-D video cubes matter

Dense workload

- 34,373 tokens → 1.181B pairs/head/block at 544P

Learned VSA route

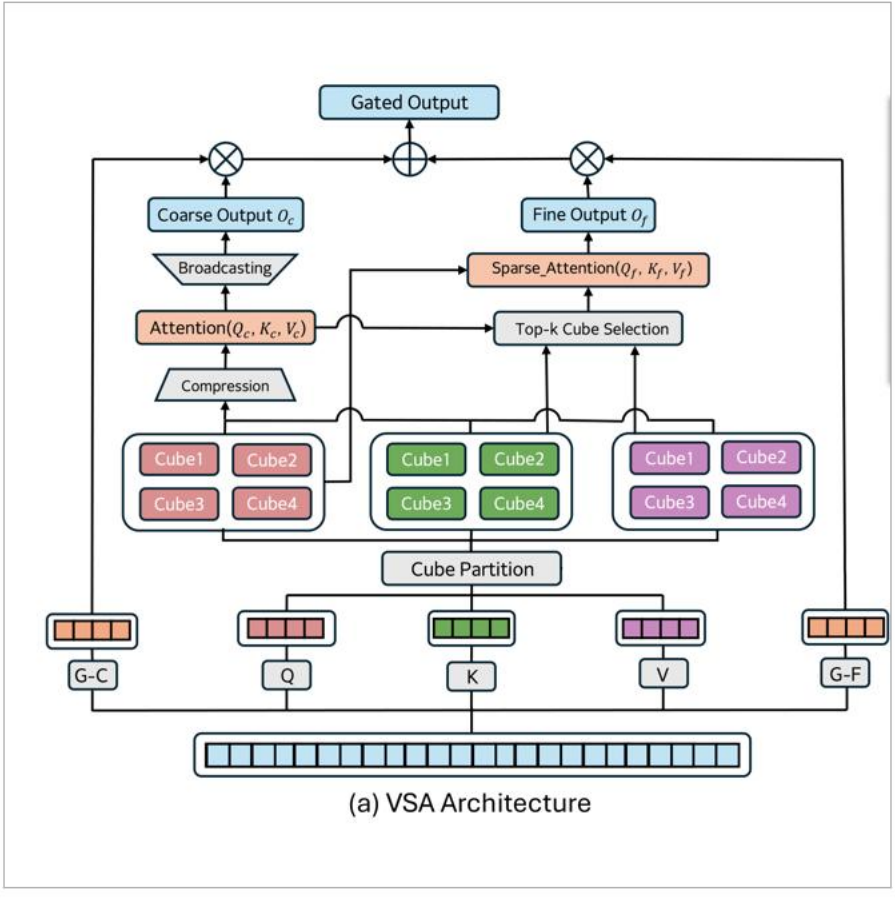
- Partition → coarse score → Top-K → fine attention → gate

Paper defaults

- 4 × 4 × 4 = 64 tokens per cube · Top-K = 32 KV cubes

System boundary

- Attention compute becomes sparse; USP8 still exchanges full-length Q/K/V/O
- Needs sparse-trained routing and gate weights



(a) VSA Architecture

Partition → Coarse → Top-K → Fine → Gate

Paper: 64-token cubes · Top-K = 32 KV cubes/query · H3: route 128 rows, kernel 64 rows

Source: Video Sparse Attention, arXiv:2505.13389, Fig. 1 · current H3 Ulysses communication path audited from source

QAD Quantizes Linears; Sage Quantizes VSA

Each path targets different SM120 kernels, and current evidence validates them separately

Path	Numerical Route	Effect	Validated Scope
QAD checkpoint	NVFP4 linears; BF16 exceptions	Recovers quantization error during distillation	8 forwards; PDD32; VSA 90%; tile 128
Sage VSA	Q/K INT8; V FP8; BF16 output	Quantizes tile64 fine attention after routing	SM120; d=128; opt-in; Ulysses
Precision island	Token refiner: BF16; VSA default: BF16	Keeps the sensitive short path in BF16	Same-seed quality check pending
Inter-GPU transport	BF16 Q/K/V before Sage; BF16 O after Sage	Keeps collective payloads in BF16	QAD and Sage do not reduce bytes

Sources: FastH3 Omni NVFP4-QAD rev 0087718 · NVIDIA ModelOpt QAD · vLLM-Omni SM120 Sage-VSA @ 63487a7

USP8 Issues Four All-to-Alls per DiT Block

Base50 runs 2,500 blocks at S = 34,432 on eight RTX PRO 5000 GPUs with local NCCL and no RDMA against a 5.167 s deadline

Scope	Tensor / Execution	A2A Rounds	Minimum Off-Rank Payload*	Rate at 5.167 s*
One DiT Block	Q / K / V forward + O reverse	4	1.728 GB	61.7 MB/rank/A2A
Base50	50 steps × 50 blocks = 2,500	10,000	4.319 TB	836.0 GB/s
PDD8	8 steps × 50 blocks = 400	1,600	691.1 GB	133.8 GB/s
Boundary	Current VSA / Sage keeps dense BF16 exchange	—	Logical bytes	Not achieved bandwidth

Source-derived @ vLLM-Omni H3 rev 63487a7f · logical one-way off-rank payload lower bound; not measured PCIe/NIC traffic or achieved bandwidth

RDMA Precheck Passes 96/96

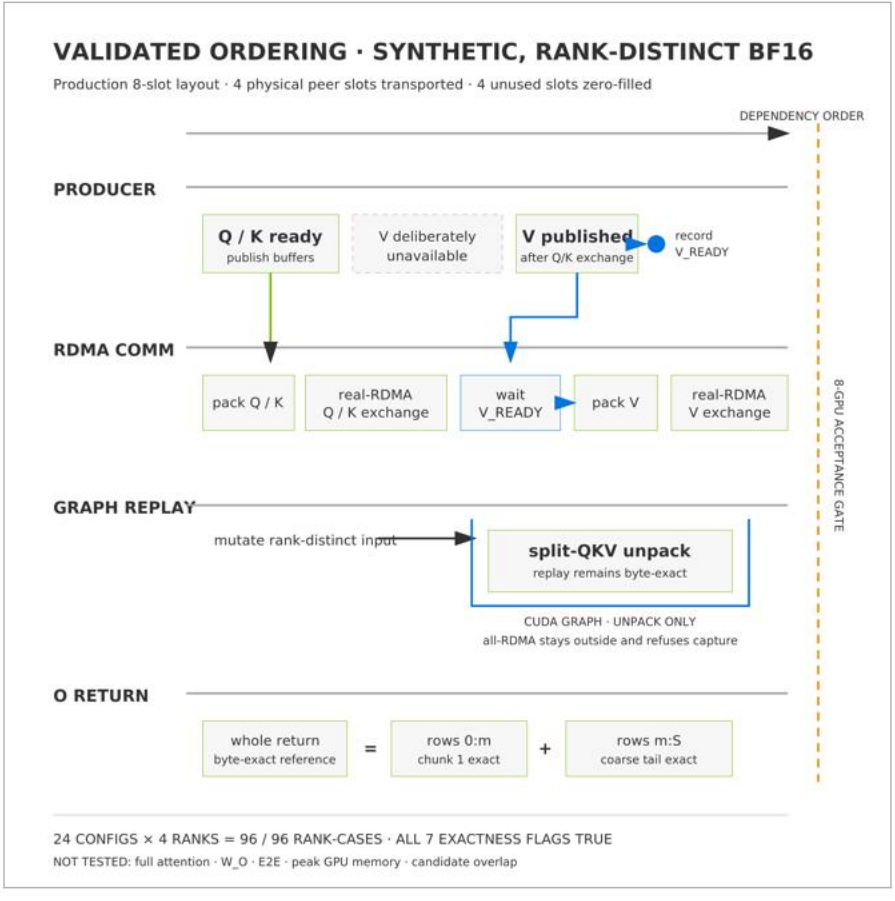
Four GPUs are byte-exact; eight-GPU H3 remains pending

Q/K first
- V waits for an explicit CUDA event

Graph replay
- Fixed-address unpack remains exact after the input changes

Tiled O return
- Whole and row-chunk returns match byte for byte

Eight-GPU acceptance gate
- Full attention + W_O, end-to-end latency, memory, and Nsight overlap



Four-GPU Real RDMA · 96 of 96 Rank Cases Exact

24 configs × 4 ranks · 8-GPU H3 candidate pending

Measured evidence: real-rdma-four-gpu-20261024-v2 · 2026-10-24 · candidate correctness precheck; latency and overlap not measured

Real-Time Demonstrated; Attribution Pending

Only Base50 uses the locked 960×544, 124-frame request; each optimized result uses a different workload

Status	Workload	System	Result / Limit
Measured baseline	960×544 · 124f · 5.167 s	Base50 · BF16 · USP8 · NCCL	247.956 s · RTF 47.992
15 s study	960×544 · 362f · 15.083 s	PDD8 · QAD · VSA · RDMA	14.974 s mean · RTF 0.993
10 s study	960×544 · 243f · 10.125 s	4-step Turbo · MXFP8 · Sage · RDMA	6.958 s · RTF 0.687
Portrait study	576×1024 · 119 new frames · 4.958 s	4-step MXFP8 · Sage · FP4 VAE · RDMA	4.459 s median · RTF 0.899
Required matched test	Lock image, audio, prompt, seed, shape, and request mode	PDD8 → VSA → QAD/Sage → RDMA	At least 3 runs per stage

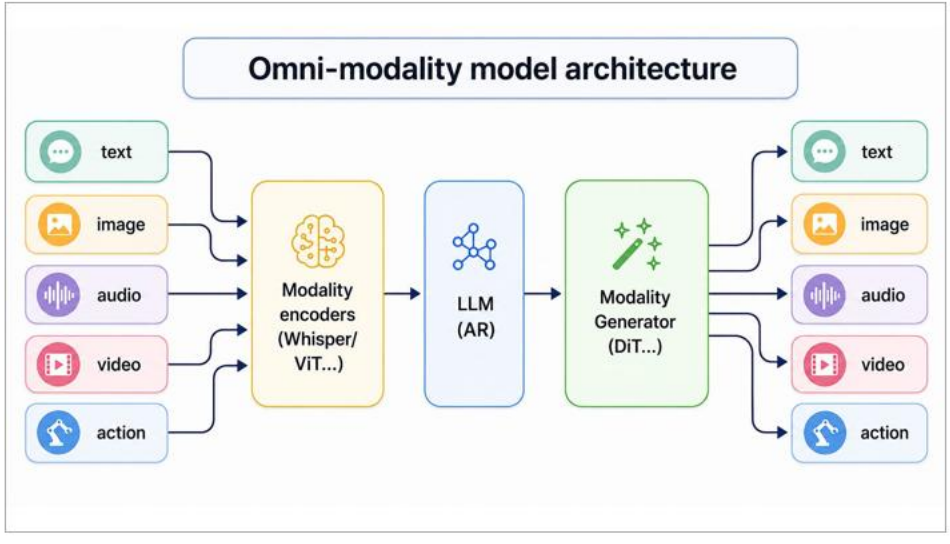
Sources: [A] Base50 r4 · [B] QAD 15 s cohort · [C] paired 10 s study · [D] portrait production · compare only within a row; timing and memory methods differ

02 · ComfyUI Apps With vLLM-Omni



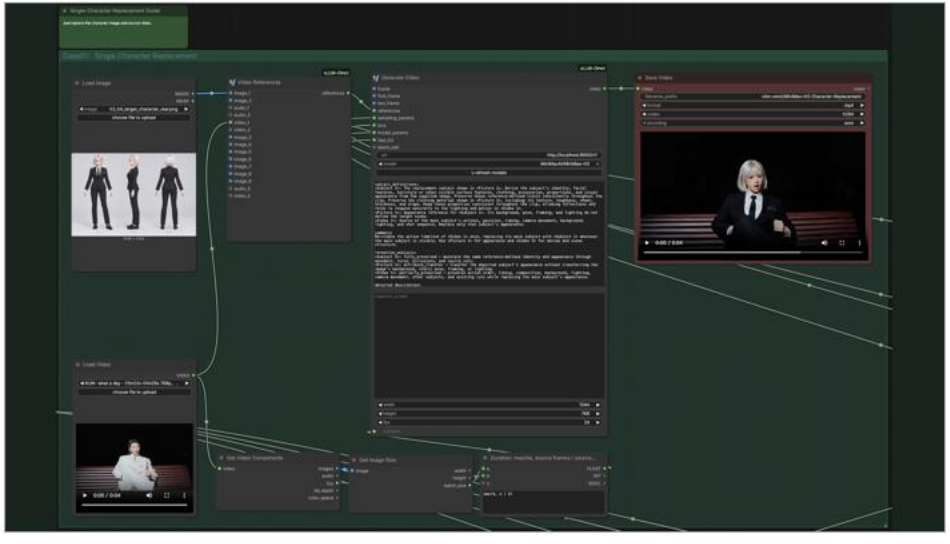
vLLM-Omni Brings Multi-GPU H3 to ComfyUI

Open PR #9194 adds character replacement, continuation-aware references, and chainable LoRA controls



Official vLLM-Omni Architecture

ComfyUI authors the workflow; vLLM-Omni runs scheduling, model execution, and multi-GPU deployment



ComfyUI Workflow · PR #9194

The workflow supports single- and multi-character replacement, source-derived duration, long-video continuation, and LoRA controls

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194 (open)

